

Specs vs Codebase Audit

Supply Chain Command Center — Updated audit after doc cleanup
Generated June 16, 2026

Executive Summary

- 82 markdown files under docs/specs/ (81 feature specs + README.md), organized across 8 domains.
- Removed since prior audit: Gen-4 roadmap, PRD folder, and README cross-domain/PRD sections.
- No broken links to removed docs remain in the repository.
- Most numbered specs are Implemented with matching backend and UI surfaces.
- Primary gaps: README index orphans, stale path references, AI Champion (backend-only), partial FVA ladder and decision-ledger policy.

What Was Removed

Removed	Impact
docs/specs/10-gen4-roadmap.md	No forward-looking backlog doc in specs
docs/specs/PRD/*	No draft/planned-work section in specs index
README cross-domain + PRD sections	Index now covers implemented features only (domains 01-08)

Spec Cleanup Still Needed

1. Orphan Specs (not in README index)

Spec	Status	Notes
02-forecasting/16-expert-system-backtest.md	Implemented	expsys_accuracy router
02-forecasting/17-ext-ml-forecast-load.md	Implemented	External ML ETL
02-forecasting/23-feature-selection-pipeline.md	Implemented	Collides with 23-lgbm-accuracy-tuning
02-forecasting/24-candidate-forecast-promotion.md	Implemented	Candidate to production promotion
04-inventory/12-service-level-unification.md	Implemented	common/core/service_levels.py
06-ai-platform/05-decision-ledger-and-policy.md	Partial	Ledger yes; policy engine no

2. Index and Numbering Issues

- Duplicate 23: 23-lgbm-accuracy-tuning (indexed) vs 23-feature-selection-pipeline (orphan).
- Forecasting index gaps: lists 15 then 18; skips 16, 17, 24.

- Demand intel gap: index 01, 03, 05-07 (no 04 file exists).
- Wrong spec ID: 07-customer-analytics.md header says Spec ID 03-04.
- Misplaced file: 01-foundation/02-sku-feature-engineering.md indexed under Demand Intelligence.

3. Stale Content Inside Specs

Spec	Problem
08-production-forecast.md	Deleted production_forecast_config.yaml; wrong script path
26-operational-reference.md	Treats algorithm_config.yaml as active
13-production-baseline-seeding.md	load_config('model_competition') example
05-champion-experimentation-studio.md	model_competition_config source
01-infrastructure.md	Says 5 champion strategies (spec 07 says 8); Chatbot + Market Intel label
07-customer-analytics.md	Status still 'Implementation'

4. Status Label Mismatches

Spec	Says	Reality
08-integration/07-fva.md	Implemented	Core tab yes; ai_adjusted / planner_adjusted planned
02-forecasting/27-ai-champion-forecast-implement	Implemented	Backend + job + SQL yes; no frontend
06-ai-platform/05-decision-ledger-and-policy.md	Policy	Policy engine removed as unwired

Feature Matrix

Legend: Implemented = full stack; Partial = scaffold or backend-only; Not built = no code

Domain	Specs	Status	Notes
01 Foundation	9	Implemented	Data layer, DQ, planning date, profiling, customer demand
02 Forecasting	29	Mostly Implemented	AI Champion (27) is backend-only
03 Demand Intelligence	5	Implemented	Clusters tab URL-only
04 Inventory	12	Implemented	Includes orphan 12-service-level-unification
05 Operations	4	Implemented	S&OP, finance, events, scenarios
06 AI Platform	5	4 Implemented, 1 Partial	Decision ledger partial
07 User Experience	8	Implemented	08-sku-validation-plan is QA playbook
08 Integration	10	9 Implemented, 1 Partial	FVA upper ladder stages planned
Removed from specs	-	Not tracked	Gen-4 roadmap and PRD removed

02 Forecasting Detail

- Core pipeline (accuracy, backtest, trees, champion, production, CI, bias, tuning studio): Implemented.
- Orphans 16, 17, 23b feature selection, 24 candidate promotion: code exists.
- Workflow docs 25, 26 and diagram 22: operational reference.
- AI Champion (27): API, job, SQL complete; no frontend tab or query module.

06 AI Platform Detail

Feature	Status	Surface
AI Planning Agent	Implemented	URL /ai-planner, not sidebar
Market Intel	Implemented	URL-only
Control Tower	Implemented	CommandCenterTab
Storyboard	Implemented	URL-only
Decision ledger + policy	Partial	Ledger only; policy not wired

Code Without Dedicated Specs

Feature	Evidence
Consensus plan + overrides	consensus_plan.py, OverrideQueuePanel, /forecast/consensus-plan
Integrated targets	integrated_targets.py

Feature	Evidence
Working capital analytics	working_capital.py, FinancialPlanPanel
Algorithm comparison	inv_planning_algorithm_comparison.py
Inv planning insights (async)	inv_planning_insights.py
Explain API	intelligence/explain.py
Accuracy budget	accuracy_budget.py
Feature lab / sampled backtest / tuning charts	no routers exist; only in tuning sub-specs
PO / sourcing, config manager, integration platform	Platform routers, no spec

Product Suggestions

Prioritized from code/spec gaps (no roadmap doc in specs).

1. README index pass

Add 6 orphan specs; renumber 23-feature-selection to 23b or 28; fix stale paths in 08-production-forecast and 26-operational-reference. Low effort, high onboarding value.

2. AI Champion UI

Biggest product gap: spec says Implemented but zero frontend usage. Surface run status, champion vs ai_champion comparison, and a generate action (Portfolio or FVA tab).

3. Override audit + planner-FVA

Consensus overrides exist; FVA planner_adjusted stage is still planned. Connect overrides to decision ledger and measured ROI.

4. Lag-decomposed accuracy API

Data already in backtest_lag_archive; no spec. High-value analytics with minimal new pipeline work.

5. Decision ledger policy engine

Rebuild against real agent write sites before expanding AI autonomy (agent_autonomy.yaml exists; engine was removed as unwired).

6. Lightweight backlog doc (optional)

If planned work should be tracked again, a single docs/specs/09-roadmap/README.md with P0/P1 bullets would fill the gap left by removing Gen-4.

Bottom Line

Metric	Count
Spec files (current)	82 (down from ~86)
Domains in index	8
Orphan specs (not indexed)	6
Stale path references	~5 specs
Backend-only vs spec	AI Champion (primary)
Draft PRD / roadmap docs	Removed

The doc trim helped: specs are closer to what ships. Remaining work is mostly index hygiene, stale path fixes, and closing backend/UI gaps (AI Champion first).